PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

1 Feb – 7 Feb 2027 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
1 Feb D2027-02-01	Daily mixed Daily test	___ / 10	Retest: 4 Feb
2 Feb D2027-02-02	Daily mixed Daily test	___ / 10	Retest: 5 Feb
3 Feb D2027-02-03	Daily mixed Daily test	___ / 10	Retest: 6 Feb
4 Feb D2027-02-04	Daily mixed Daily test	___ / 10	Retest: 7 Feb
5 Feb D2027-02-05	Daily mixed Daily test	___ / 10	Retest: 8 Feb
6 Feb D2027-02-06	Daily mixed Daily test	___ / 10	Retest: 9 Feb
7 Feb T2027-02-07	SST Full paper (3 hr)	___ / 80	Retest: 10 Feb

Daily Test · Answer key

Monday, 1 February 2027 · 10 marks

Test ID: D2027-02-01

SST Ch 8 Problem of Choice · Science Ch 8 Atom · Maths Ch 8 Sequences & Progressions

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Q1 [2] **Question:** Why does the problem of choice arise?

Expected answer / marking points: Human wants are unlimited [1] but resources are limited (scarce) and have alternative uses [1].

Q2 [2] **Question:** Differentiate between a market economy and a planned economy.

Expected answer / marking points: Market: decisions by private individuals through demand and supply [1]; planned: decisions by the government [1].

Q3 [1] **Question:** Who discovered the electron?

Expected answer / marking points: J. J. Thomson.

Q4 [2] **Question:** An atom has 11 protons and 12 neutrons. Find its atomic number, mass number and electronic configuration.

Expected answer / marking points: Z = 11, A = 23 [1]; 2, 8, 1 [1].

Q5 [3] **Question:** A ball is dropped from 81 m and each bounce reaches 2/3 of the previous height. Find the heights of the first three bounces and the height of the 5th bounce.

Expected answer / marking points: 54, 36, 24 m [1.5]; 5th = $81 \times (2/3)^5 = 81 \times 32/243 = 32/3 \approx 10.67$ m [1.5].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Tuesday, 2 February 2027 · 10 marks

Test ID: D2027-02-02

Maths Ch 9 Propositions · AI Data Literacy · English Unit 7

Q1 [1]

Question: Is “Please close the door.” a proposition? Why?

Expected answer / marking points: No — it is a request; it cannot be true or false.

Q2 [2]

Question: Write the converse of “If a number ends in 0, then it is divisible by 5.” Is it true?

Expected answer / marking points: If a number is divisible by 5, then it ends in 0 [1]; false — 15 is divisible by 5 but ends in 5 [1].

Q3 [3]

Question: Explain the steps of the data processing cycle.

Expected answer / marking points: Collection → preparation/cleaning → input → processing → output/interpretation → storage [3].

Q4 [2]

Question: Why does Khetaram stand outside the house to read a letter with bad news?

Expected answer / marking points: He believes “Ashubh Samachar cannot be carried into the house” [1]; he reads it twice and tears it, as bad news must be destroyed [1].

Q5 [2]

Question: How was the purpose of India Post different from the British postal system?

Expected answer / marking points: The British system mainly carried company dak between administrative centres [1]; after Independence, India Post aimed to bring the whole population within reach of the mail [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Wednesday, 3 February 2027 · 10 marks

Test ID: D2027-02-03

Hindi पाठ 9 राम–लक्ष्मण–परशुराम संवाद · SST Ch 9 Price Puzzle · Science Ch 9 Atomic Foundations

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Q1 [1]

Question: इस अंश की भाषा कौन-सी है?

Expected answer / marking points: अवधी।

Q2 [2]

Question: लक्ष्मण ने धनुष टूटने पर परशुराम से क्या तर्क दिया?

Expected answer / marking points: बचपन में हमने बहुत-सी धनुहियाँ तोड़ीं, तब आप क्रोधित नहीं हुए [1]; यह पुराना धनुष राम के छूते ही टूट गया, इसमें उनका क्या दोष [1]।

Q3 [1]

Question: Tea and coffee are substitutes or complements?

Expected answer / marking points: Substitutes.

Q4 [2]

Question: Give two factors (other than price) that affect demand.

Expected answer / marking points: Income, prices of related goods, tastes, season, expectations, population (any two) [2].

Q5 [2]

Question: How is an ionic bond formed in sodium chloride?

Expected answer / marking points: Na (2, 8, 1) gives 1 electron to become Na^+ [1]; Cl (2, 8, 7) takes it to become Cl^-; the opposite ions attract — ionic bond [1].

Q6 [2]

Question: Write two properties of ionic compounds.

Expected answer / marking points: Any two: high melting/boiling points, conduct electricity in molten state or solution, usually soluble in water, hard solids [2].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Thursday, 4 February 2027 · 10 marks

Test ID: D2027-02-04

SST P2 Ch 1 Oceans · Science Ch 10 Sound · Maths Ch 10 Data

Q1 **[2]**

Question: Differentiate between waves and ocean currents.

Expected answer / marking points: Waves: up-and-down movement of the water surface caused mainly by wind [1]; currents: large streams of water moving steadily in a direction over long distances [1].

Q2 **[2]**

Question: What rights does a country have in its EEZ?

Expected answer / marking points: Exclusive rights to explore and use natural resources — fish, oil, gas, minerals [1]; other countries may still navigate/fly over [1].

Q3 **[1]**

Question: Sound waves in air are longitudinal or transverse?

Expected answer / marking points: Longitudinal.

Q4 **[3]**

Question: A person claps near a cliff and hears the echo after 2 s. If the speed of sound is 340 m s^{-1}, how far is the cliff? Why do concert halls use soft materials on walls?

Expected answer / marking points: d = 340 × 2/2 = 340 m [1.5]; soft materials absorb sound and reduce reverberation [1.5].

Q5 **[2]**

Question: The mean of 5 numbers is 18. One number, 30, is removed. Find the mean of the remaining numbers.

Expected answer / marking points: Total 90 [0.5]; 90 – 30 = 60 [0.5]; 60/4 = 15 [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Friday, 5 February 2027 · 10 marks

Test ID: D2027-02-05

Maths Ch 11 Algorithms · AI Generative AI · English Grammar + Writing

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Q1 [3]

Question: Write an algorithm (in steps) to find the sum of all even numbers from 1 to n. Trace it for n = 7.

Expected answer / marking points: sum = 0; for each j from 1 to n: if j is even, add j to sum; output sum [2]. n = 7: 2 + 4 + 6 = 12 [1].

Q2 [1]

Question: For a list of 10 numbers, how many comparisons does "compare each number with the current maximum" make to find the largest?

Expected answer / marking points: 9 comparisons.

Q3 [1]

Question: Give one example of a Generative AI tool.

Expected answer / marking points: Any one: a chatbot that writes text, an image-generator, a music generator.

Q4 [2]

Question: Give two benefits of Generative AI for students.

Expected answer / marking points: Explains topics, helps brainstorm ideas, gives practice questions (any two) [2].

Q5 [3]

Question: Write a diary entry (60–80 words) about the day you helped an elderly neighbour.

Expected answer / marking points: Format: date/day/time + "Dear Diary" [0.5]; content — what happened, how you helped, feelings [2]; expression/sign-off [0.5].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Saturday, 6 February 2027 · 10 marks

Test ID: D2027-02-06

Hindi पाठ 11 झाँसी की रानी · SST P2 Ch 2 Life on Earth · Science Ch 11 Reproduction

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Q1 [1]

Question: रानी किस वर्ष के स्वतंत्रता संग्राम से जुड़ी हैं?

Expected answer / marking points: 1857।

Q2 [3]

Question: इस कविता से हमें कौन-सी तीन प्रेरणाएँ मिलती हैं?

Expected answer / marking points: देश की स्वतंत्रता के लिए बलिदान [1]; स्त्रियाँ भी पुरुषों के समान वीर होती हैं [1]; अन्याय के आगे न झुकना [1]।

Q3 [2]

Question: Why does India have diverse biomes?

Expected answer / marking points: Large size and range of latitude [1]; varied relief, rainfall and temperature — mountains, deserts, coasts, plains [1].

Q4 [2]

Question: Name two occupations linked with forests.

Expected answer / marking points: Collecting minor forest produce (honey, tendu leaves), timber/forestry, herbal medicine, ecotourism (any two) [2].

Q5 [2]

Question: Give two advantages of vegetative propagation.

Expected answer / marking points: Plants identical to parent, quicker flowering/fruiting, useful for plants without viable seeds (e.g., banana) — any two [2].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Social Science · Full Paper 1 (Annual pattern) · Answer key

Sunday, 7 February 2027 · 80 marks

Test ID: T2027-02-07

Class 9 · Understanding Society: India and Beyond — Part 1 (Ch 1–9) + Part 2 (Ch 1–6)

Q1 [1]

Question: Which discipline of Social Science deals with the use of scarce resources?

(a) History
(b) Geography
(c) Political Science
(d) Economics

Expected answer / marking points: (d) Economics

Q2 [1]

Question: The innermost layer of the Earth is the:

(a) crust
(b) mantle
(c) core
(d) lithosphere

Expected answer / marking points: (c) core

Q3 [1]

Question: Sand dunes are formed mainly by:

(a) running water
(b) glaciers
(c) wind
(d) sea waves

Expected answer / marking points: (c) wind

Q4 [1]

Question: The layer of the atmosphere in which aeroplanes usually fly is the:

(a) troposphere
(b) lower stratosphere
(c) mesosphere
(d) exosphere

Expected answer / marking points: (b) lower stratosphere

Q5 [1]

Question: The India Meteorological Department divides the year into how many seasons?

(a) Two
(b) Three
(c) Four
(d) Six

Expected answer / marking points: (c) Four

Q6 [1]

Question: Use of copper along with stone tools marks the:

(a) Palaeolithic period
(b) Mesolithic period
(c) Chalcolithic period
(d) Iron Age

Expected answer / marking points: (c) Chalcolithic period

Q7 [1]

Question: Which text on statecraft is associated with Kauṭilya?

(a) Ṛigveda
(b) Arthaśhāstra
(c) Mahābhārata
(d) Indika

Expected answer / marking points: (b) Arthaśhāstra

Q8 [1] **Question:** Universal adult franchise means:

(a) only educated people vote
(b) every adult citizen can vote
(c) only taxpayers vote
(d) only men vote

Expected answer / marking points: (b) every adult citizen can vote

Q9 [1] **Question:** The body that fixes the boundaries of constituencies is the:

(a) Election Commission
(b) Delimitation Commission
(c) Supreme Court
(d) Finance Commission

Expected answer / marking points: (b) Delimitation Commission

Q10 [1] **Question:** India is an example of a:

(a) pure market economy
(b) planned economy
(c) mixed economy
(d) barter economy

Expected answer / marking points: (c) mixed economy

Q11 [1] **Question:** Tea and coffee are:

(a) complementary goods
(b) substitute goods
(c) public goods
(d) inferior goods

Expected answer / marking points: (b) substitute goods

Q12 [1] **Question:** Under UNCLOS, a country's territorial sea extends up to:

(a) 3 nautical miles
(b) 12 nautical miles
(c) 24 nautical miles
(d) 200 nautical miles

Expected answer / marking points: (b) 12 nautical miles

Q13 [1] **Question:** A large region with similar climate, plants and animals is called a:

(a) delta
(b) biome
(c) plateau
(d) basin

Expected answer / marking points: (b) biome

Q14 [1] **Question:** The Vijayanagara Empire was founded in:

(a) 1206
(b) 1336
(c) 1526
(d) 1674

Expected answer / marking points: (b) 1336

Q15 [1] **Question:** Who established the Khalsa in 1699?

(a) Guru Nanak
(b) Guru Arjan Dev
(c) Guru Tegh Bahadur
(d) Guru Gobind Singh

Expected answer / marking points: (d) Guru Gobind Singh

Q16 [1] **Question:** Which is NOT a principle of good governance?

(a) Transparency (b) Accountability

(c) Secrecy in all decisions (d) Rule of law

Expected answer / marking points: (c) Secrecy in all decisions

Q17 [1] **Question:** A plan of expected income and expenses is a:

(a) budget (b) loan

(c) tax (d) premium

Expected answer / marking points: (a) budget

Q18 [1] **Question:** Rising prices that reduce the value of money is called:

(a) deflation (b) inflation

(c) interest (d) profit

Expected answer / marking points: (b) inflation

Q19 [1] **Question:** Revenue minus expenses gives:

(a) cash flow (b) profit

(c) capital (d) turnover

Expected answer / marking points: (b) profit

Q20 [1] **Question:** Which functional area of a business deals with hiring and training staff?

(a) Marketing (b) Finance

(c) Human resources (d) Operations

Expected answer / marking points: (c) Human resources

Q21 [2] **Question:** Differentiate between weather and climate.

Expected answer / marking points: Weather: day-to-day atmospheric conditions [1]; climate: average weather of a place over a long period (about 30 years) [1].

Q22 [2] **Question:** Give two features of the Mesolithic period.

Expected answer / marking points: Microliths (small stone tools); temporary settlements near rivers/lakes; hunting-gathering with some domestication (any two) [2].

Q23 [2] **Question:** What is the Model Code of Conduct? Give one example of a rule under it.

Expected answer / marking points: Guidelines for parties and candidates during elections [1]; e.g., no use of government vehicles/resources for campaigning; no appeal on caste/religion [1].

Q24 [2] **Question:** What rights does India have in its Exclusive Economic Zone?

Expected answer / marking points: Exclusive right to explore and use resources — fish, oil, gas, minerals [1]; up to 200 nautical miles; other ships can still pass [1].

Q25 [3] **Question:** How do plate movements explain the distribution of earthquakes and volcanoes?

Expected answer / marking points: The crust is broken into plates that move due to heat in the interior [1]; most earthquakes and volcanoes occur along plate boundaries where plates collide, separate or slide [1]; e.g., Himalayas/Pacific Ring of Fire [1].

Q26 [3] **Question:** Explain three ways in which the Mauryan Empire was administered.

Expected answer / marking points: Emperor at the top with council of ministers [1]; provinces under princes/governors, officials for revenue, roads, markets [1]; spies, army and taxation system (as described in the Arthaśhāstra) [1].

Q27 [3] **Question:** Elections are called the soul of democracy. Give three reasons.

Expected answer / marking points: People choose representatives [1]; equal vote, every citizen's voice counts [1]; governments can be changed peacefully, which keeps them accountable [1].

Q28 [3] **Question:** Explain the three central problems of an economy.

Expected answer / marking points: What to produce (which goods, how much) [1]; how to produce (labour- or machine-intensive) [1]; for whom to produce (distribution) [1].

Q29 [3] **Question:** Explain three ways in which forest dwellers and traditional communities help conserve forests.

Expected answer / marking points: Sacred groves / traditional beliefs protect trees [1]; sustainable collection of forest produce [1]; guarding against fires and illegal cutting; movements like Chipko/Bishnoi [1].

Q30 [5] **Question:** Describe the features of the Palaeolithic, Mesolithic and Neolithic periods in terms of tools, settlement and way of life.

ATHVA / OR

Describe the main features of any two early world civilisations (Sindhu–Sarasvatī, Mesopotamia, Egypt, China).

Expected answer / marking points: Palaeolithic: large stone tools (handaxes, cleavers), caves/open camps, hunter-gatherers [1.5]; Mesolithic: microliths, temporary settlements near water, hunting + gathering [1.5]; Neolithic: polished tools, farming, domestication, permanent villages, pottery, weaving [2]. **OR:** Any two, each with river setting, cities/architecture, writing, trade/agriculture, administration — e.g., Sindhu–Sarasvatī: planned cities, drainage, seals, trade; Egypt: Nile, pyramids, hieroglyphs, pharaoh [2.5 + 2.5].

Q31 [5] **Question:** Explain any five principles on which Indian democracy is based according to the Constitution.

Expected answer / marking points: Popular sovereignty; universal adult franchise; fundamental rights and duties; rule of law; separation of powers (and accountability/equality) — each explained [1 × 5].

Q32 [5] **Question:** State the laws of demand and supply. Explain, with a diagram, how market equilibrium is reached.

ATHVA / OR

Explain the five pillars of personal finance. Why is it important to start managing money at a young age?

Expected answer / marking points: Law of demand [1]; law of supply [1]; diagram with D and S curves crossing at E [1]; above E surplus → price falls; below E shortage → price rises; equilibrium where Qd = Qs [2].
OR: Income, budgeting, saving, investing, protection (insurance) — each explained [3.5]; early habits build security, independence and the power of compounding/meeting goals [1.5].

Q33 [5] **Question:** What is good governance? Explain five principles of good governance with examples.

Expected answer / marking points: Definition [1]; participation, transparency, accountability, rule of law, responsiveness/efficiency — each with a brief example [4].

Q34 [4] **Question:**

> During the monsoon, a village on a hill slope faced landslides after many trees were cut for a new road. The gram sabha decided to plant native trees, build small check dams and stop cutting on steep slopes.

(i) Which agent of gradation was most active here? [1]
(ii) How is deforestation linked to soil erosion? [1]
(iii) Suggest two more measures to protect land from erosion in hilly areas. [2]

Expected answer / marking points: (i) Running water (rain) [1]. (ii) Roots hold soil; without trees rain washes soil away [1]. (iii) Terrace farming, contour ploughing, grass cover, retaining walls, controlled construction (any two) [2].

Q35 [4] **Question:**

> In 1931, two brothers started a small business in Mumbai selling ink powder, using recycled glass bottles. They faced money problems and found it hard to win customers' trust. They focused on good quality at fair prices, noticed that Indian artists lacked good art materials, travelled abroad to study colour chemistry, and set up a laboratory. The company grew into a well-known stationery brand.

(i) Name one entrepreneurial quality the brothers showed. [1]
(ii) What opportunity did they identify? [1]
(iii) Which functional area helped them improve their products, and how did their business serve society apart from earning profit? [2]

Expected answer / marking points: (i) Persistence / risk-taking / innovation / customer focus [1]. (ii) Lack of good, affordable art materials for Indian artists [1]. (iii) Operations/R&D (laboratory, colour chemistry) [1]; provided quality, affordable stationery, created jobs, supported artists/students [1].

Q36 [4] **Question:**

> A district had low use of government health services. The administration ran awareness camps, set up a helpline for complaints, published monthly reports of medicines available in each health centre, and asked village health committees to give feedback.

(i) Which public problem was being addressed? [1]
(ii) Which principle of good governance is shown by publishing monthly reports? [1]
(iii) Name two actors other than the government involved here and the role of citizens in the policy's success. [2]

Expected answer / marking points: (i) Low use/poor reach of public health services [1]. (ii) Transparency (also accountability) [1]. (iii) Village health committees/civil society, media, citizens [1]; citizens give feedback, use services, raise complaints [1].

Q37 [5] **Question:** On an outline political map of India, locate and label any five: (a) the Himalayas (b) the Thar desert (c) the Arabian Sea (d) the Lakshadweep Islands (e) Hampi (f) Haldighati (Rajasthan) (g) a coastal state that may need tsunami warnings.

Expected answer / marking points: Each correctly located and labelled [1 × 5]: Himalayas — northern border; Thar — western Rajasthan; Arabian Sea — west coast; Lakshadweep — off Kerala coast; Hampi — Karnataka; Haldighati — near Udaipur, Rajasthan; coastal state — e.g., Tamil Nadu/Odisha/Andhra Pradesh/Kerala.

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: